KLinterSel Manual v 0.3.1

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Versions

Version KLinterSel 0.3.1 (October 2026):

- The HGkI test is now the default analysis method.

- Improved computational efficiency in several core routines.

- iHDSel output files containing significant candidate positions (*_SIG.tsv) can now be used directly as input files in KLinterSel.

- KLinterSel now supports .norm output files from both selscan 2.1 and 3.x, including formats with or without column headers.

- Added validation of the original SNP-position file format.

- Added consistency checks between original SNP positions and candidate files.

- Added safeguards against using single-chromosome .norm or *_SIG.tsv files with multi-chromosome datasets.

- Improved the --stats option to provide more informative summaries of input files, including the total number of analyzed sites in the original data file and the number of candidate sites detected by each method.

- Added per-chromosome reporting and safe handling of chromosomes or candidate files containing no positions.

- When cluster filtering is applied, --stats reports its effect on the number of candidates, facilitating the selection of appropriate --Kmax and --dist values before running the analyses.

Version KLinterSel 0.3 (December 2025 - June 2026):

- A new parametric test (HGkI) based on a sequentially conditioned hypergeometric distribution.

- A rank-based measure to assess deviations from uniformity in the spatial distribution of SNPs along the chromosome.

Version KLinterSel 0.2 (November 2025):

- Major speed-up of the KL computation thanks to a more efficient RelEntr implementation.

- New command-line options:

    --permissive to disable the strict redundancy filter between candidate files.

    --uniform perform resampling assuming a uniform distribution of SNPs.

- Sorted intersection output: elements within each intersection are now returned in sorted order for consistent and cleaner reporting.

- Bug fix: corrected an issue in the resampling test where the comparison of medians introduced a slight conservative bias.

Version KLinterSel 0.1 (August 2025):

- Improvements for figure customization. Multiprocessor options.

Version KLinterSel 0.0 (July 2025):

- This is the first version.

Introduction

KLinterSel is a Python project designed to assess the agreement among different methods for detecting candidate sites under selection. It implements a Monte Carlo–based test (T_{KL}) and a hypergeometric k-way intersection test (HGkI) to evaluate whether observed overlaps are closer than expected by chance. The program also computes intersections among candidates and provides complementary plotting and statistical summaries.

Pre-built Binaries Requiring No Installation

Pre-built KLinterSel binaries are available at https://github.com/noosdev0/KLinterSel/releases/tag/v0.3.1

Binaries are provided for Windows, Linux, and macOS (arm64) and should work on most versions of these operating systems. To run them, open a terminal or command prompt in the folder containing the KLinterSel executable and the input files (e.g., totsnps.txt, sigsmethod1.norm, sigsmethod2.tsv, and sigsmethod3.norm). These filenames are examples and can be replaced with the actual names of your files.

Starting with version 0.3.1, KLinterSel runs the hypergeometric HGkI test by default. To run the Monte Carlo T_{KL} test instead, simply add the --TKL flag.

Linux:

```
./KLinterSel totsnps.txt sigsmethod1.norm sigsmethod2.tsv sigsmethod3.norm --path .
```

macOS:

```
./KLinterSel totsnps.txt sigsmethod1.norm sigsmethod2.tsv sigsmethod3.norm --path .
```

Windows:

Double-click the KLinterSel executable to run the program with the default options. The program will prompt you to enter the required filenames. Alternatively, open the command prompt (cmd.exe), navigate to the KLinterSel folder, and type:

```
KLinterSel.exe totsnps.txt sigsmethod1.norm sigsmethod2.tsv sigsmethod3.norm
--path .
```

Additional options can be specified as needed, such as --TKL to run the Monte Carlo test, --perm 10000 to set the number of resampling iterations, or --dist 10000 to define a distance threshold of 10 kb.

In windows you can also access the Run command by pressing the Windows logo key +r then drag and drop the .exe file from your folder and add the desired arguments, e.g.:

```
C:\KLinterSel\KLinterSel_Win.exe totsnps.txt sigsmethod1.norm sigsmethod2.tsv
sigsmethod3.norm --path . --perm 1000 --dist 10
```

Installation

Clone the KLinterSel repository or download the files. To use the KLinterSel script (KLinterSel.py), you need to have Python installed (version 3.7 or higher). To install the necessary dependencies, navigate to the folder containing the KLinterSel.py script where the requirements.txt file should also be located, and run the following command in the terminal:

```
pip install -r requirements.txt
```

This command will install all required libraries as specified in the requirements.txt file, including numpy, pandas, matplotlib, seaborn, scipy and psutil, along with any other dependencies listed.

To run the Python script, open a terminal in the folder containing KLinterSel.py and the input files (e.g., totsnps.txt, sigsmethod1.norm, sigsmethod2.tsv, and sigsmethod3.norm), and type:

```
python3 KLinterSel.py totsnps.txt sigsmethod1.norm sigsmethod2.tsv
sigsmethod3.norm
```

Input (Data Format)

The script requires two types of input files:

1.- Original Positions Data File: The first input file contains the positions of all analyzed SNPs. This file can be provided in PLINK MAP (.map) format, consisting of four columns without a

header: chromosome, SNP identifier, genetic position, and physical position (in base pairs). KLinterSel uses the chromosome and physical position columns.

Alternatively, CSV (.csv), TSV (.tsv), or plain-text (.txt) files are also supported, with the following structure:

```
CHR POS
1 12345
1 12367
.
2 3402
.
.
```

The first column identifies the chromosome, and the second column lists the position of the SNP within the chromosome.

2.- Candidate Site Results Files: Files containing the candidate positions identified by each method. These files can have the same two-column format as the original data file (CSV, TSV, or TXT). Normalized output files (.norm) from selscan versions 2.1 and 3.x are supported. Candidate positions are identified using the pos column and selected when the value in the crit column is equal to or greater than 1.

In addition, iHDSel output files containing significant candidates (*_SIG.tsv) can be used directly as input files, with candidate positions read from the column named SNP.

KLinterSel currently processes .norm and *_SIG.tsv files as single-chromosome datasets, internally assigning all candidate positions to chromosome 1. Although .norm files generated by selscan 3.x include a chromosome identifier, this information is currently ignored.

Important: When one or more .norm or *_SIG.tsv files are included in an analysis, all input files must contain data for a single chromosome. In all other input formats that explicitly include a chromosome identifier (e.g., .map, .txt, .tsv, .csv, .tped, or .hapflk), that chromosome must be identified as 1. Thus, when analysing a biological chromosome other than chromosome 1, its chromosome identifier must be recoded as 1 in these input files. For example, when analysing human chromosome

2 together with .norm files, the first column of the corresponding .map or other chromosome-labelled files must contain 1, not 2. This recoding affects only the chromosome identifier used internally by KLinterSel; physical SNP positions must remain unchanged.

Command Line Arguments

A full description of all available command-line options can be displayed by running:

```
python3 KLinterSel_v0.3.1.py --help
```

Below is a breakdown of the various command line arguments that can be used with the script:

- **Version**

  -v, --version Display the current version of KLinterSel and exit.

- **Basic Command**

  python3 KLinterSel_v0.3.1.py totsnps.txt sigsmethod1.norm sigsmethod2.tsv sigsmethod3.norm

  This command processes the data without additional options it is equivalent to

  python3 KLinterSel_v0.3.1.py totsnps.txt sigsmethod1.norm sigsmethod2.tsv sigsmethod3.norm --path . --HG --SL 0.05.

- **Distance option**

  --dist 10000 Specify the distance threshold in base pairs (default: 10,000) for computing intersections and, in combination with --Kmax, for grouping nearby candidate sites when the number of candidates exceeds the --Kmax threshold.

- **Hypergeometric k-way intersection test (HGkI)**

  Starting with version 0.3.1, HGkI is the default statistical test.

--HG Explicitly selects the hypergeometric HGkI test. This option is retained for backward compatibility, as HGkI is now the default test.

--W Specifies the size of the non-overlapping genomic units used by HGkI, independently of the distance threshold defined by --dist. When W = 1 (default), the units correspond to individual SNP positions (exact coincidences). When W > 1, the units correspond to non-overlapping genomic windows of size W base pairs containing at least one SNP. Multiple candidate SNPs falling within the same window are counted as a single event.

- **Kullback–Leibler-like Monte Carlo test (T_{KL})**

  --TKL Enables the distance-based T_{KL} test instead of the default HGkI test.

- **Intersections**

  --intersection Enables the computation of intersections among candidate positions from different methods, which is disabled by default. When a statistical test (HGkI or T_{KL}) is performed, intersections are computed only for chromosomes identified as significant. To compute intersections for all chromosomes without performing a statistical test, use --intersection --notest.

  Alternatively, --intersection --SL 1 performs the selected statistical test but allows intersections to be reported regardless of statistical significance.

- **Chromosome selection (HGkI, T_{KL}, and intersections)**

  --chr-id N Restricts the statistical analysis and, when --intersection is enabled, the computation of intersections to chromosome N, where N is an integer from 1 to the total number of chromosomes in the dataset. By default, all chromosomes are analyzed.

  This option applies to both HGkI and T_{KL} and can also be combined with --intersection --notest to identify

intersections only in the selected chromosome without performing statistical tests. When a statistical test is performed, intersections for the selected chromosome are reported only if that chromosome is significant.

- **Kmax option**

  --Kmax undefined If defined and the number of candidate SNPs in any file is greater than Kmax, it filters SNP positions to group those that are at distance ≤ *D*, leaving only the first and last of each group.

- **No test option**

  --notest No statistical tests (T_{KL} or HGkI) are performed.

- **Number of permutations**

  --perm 10000 Specifies the number of Monte Carlo resampling iterations used to compute the expected distance profile (default: 10,000). This option applies only when --TKL is enabled.

- **Path Specification**

  --path PATH Specifies the directory containing the input files (default: current working directory, .).

- **Permissive with redundant methods**

  --permissive Disables the default rule that removes candidate files showing high redundancy (i.e., when all or most candidate sites are identical across methods).

- **Plotting**

  --paint This option applies only when --TKL is enabled. Enable the generation of plots. In this case, intersection computations are skipped. The program draws two histograms: the one on the left shows the observed distances, and the one on the right shows the expected distance profile if the matches between methods were

generated at random given the genomic locations of SNPs in the input data. If there are multiple chromosomes, the program displays the graph for each chromosome and, after closing it, continues with the next chromosome. If you only want a graph for a specific chromosome, for example chromosome 7, you can specify --paint 7.

The scale of the X and Y axes can be controlled with the --max-xvalue and --max-yvalue arguments, respectively. The first is the Mb scale (--max-xvalue 20) and the second is a frequency between 0 and 1 (--max-yvalue 0.25).

- **Significance level**

  --SL 0.05 Set the significance level for the statistical tests (default 0.05).

- **Stats**

  --stats Generates summary statistics for each chromosome in each input file, including minimum, maximum, range, mean, standard deviation, and median values, together with the total number of analyzed sites in the original file and the number of candidate sites in each candidate file. If cluster filtering is applied, the effect of the --Kmax and --dist settings on the number of retained candidates is also reported. This option is useful for inspecting the input data and, in particular, for deciding whether candidate filtering with --Kmax and --dist may be advisable before running T_{KL}. No other calculations are performed in this case.

- **Uniform**

  --uniform For the T_{KL} test, perform resampling assuming a uniform SNP distribution.

Output

The KLinterSel script generates three types of output files.

HGkI Test Results Output File (HGkItest_)

- This file reports the results of the HGkI test for each chromosome. It includes the total number of SNPs and occupied windows, the number of candidate SNPs and windows identified by each method, the observed number of overlapping windows (k_{obs}), the number expected under the null model E[k_{obs}], the associated p-value, and a rank-based measure of deviation from uniformity in the spatial distribution of SNPs along the chromosome (0 = lowest rank of deviation, 1 = highest rank of deviation), together with an independent, test-based classification derived from Kolmogorov–Smirnov, Cramér–von Mises, and cv_counts (coefficient of variation of SNP counts) statistics (close to uniform, medium deviation, or strong deviation).

  Under the null hypothesis of independent and uniformly distributed candidates across windows, the expected number of k-way overlapping windows is

$$E[k_{obs}] = \frac{\prod_{i=1}^{k} n_i}{N^{k-1}}$$

  where n_i is the number of windows occupied by method i and N is the total number of windows in the given chromosome. This expectation ranges from 0 to the smallest number of windows reported by any method.

T_{KL} Test Results Output File (TKLtest_)

- This file reports the results of the T_{KL} test for each chromosome. It includes the total number of SNPs and the number of candidate SNPs identified by each method, a Kullback-Leibler-like discrepancy measure between the observed and expected ordered distance profiles, the associated p-value, the observed (oQ2) and expected (eQ2) medians, and a rank-based measure of deviation from uniformity in the spatial distribution of SNPs along the chromosome (0 = lowest rank of deviation, 1 = highest rank of deviation), together with an independent, test-based

classification derived from Kolmogorov–Smirnov, Cramér–von Mises, and cv_counts (coefficient of variation of SNP counts) statistics (close to uniform, medium deviation, or strong deviation).

Intersections Result File (INTERSEC_<Test name>_D1E4_)

- This file contains, for each chromosome, the intersections among the different methods. The maximum distance threshold used to define intersections is indicated in the file name; for example, _D1E4_ denotes a threshold of 10 kb, meaning that candidate positions separated by 10 kb or less are considered for intersection.
- Intersections are reported for all combinations of methods. For instance, when three methods are used (K = 3), both the pairwise intersections and the three-way intersection are provided. The last column lists the genomic positions involved in the intersection across all methods.

Plots (if enabled)

- If the plotting option is enabled, the program generates graphical representations of the observed and expected distance profiles that can be saved to the user's computer.

Spatial uniformity ranking metric

In the test results output files of both HGkI and T_{KL}, the last columns report a rank-based measure of deviation from spatial uniformity together with a test-based classification derived from the Kolmogorov–Smirnov, Cramér–von Mises, and cv_counts statistics (coefficient of variation of SNP counts). The classification labels chromosomes as close to uniform, moderately deviating, or strongly deviating from uniformity.

This information is provided because both HGkI and T_{KL} explicitly account for the empirical spatial distribution of SNPs, making it useful to characterize the degree of spatial regularity of

SNP positions across chromosomes. To facilitate this interpretation,the program computes a spatial uniformity ranking metric, which orders chromosomes according to the extent to which their SNP distributions depart from uniformity.

Chromosomes are ranked on a normalized scale from 0 to 1, where 0 indicates spatial uniformity and 1 indicates extreme deviation from uniformity. The ranking is based on a composite deviation score defined as the mean percentile rank across three complementary statistics: Kolmogorov–Smirnov (KS), Cramér–von Mises (CvM), and binned count heterogeneity (measured as the coefficient of variation of SNP counts across genomic bins). This procedure yields a scale-independent multivariate summary of spatial deviation from uniformity. In the program output, the ranking score and the associated classification are reported under the headings URank and Uclasf, respectively.

The composite score emphasizes consistent deviations across multiple measures of spatial heterogeneity. Each statistic captures a different aspect of spatial structure. The Kolmogorov–Smirnov (KS) statistic is sensitive to localized departures from uniformity, highlighting sharp clustering or large gaps. The Cramér–von Mises (CvM) statistic summarizes cumulative deviations across the entire chromosome and is therefore more sensitive to broad-scale or distributed departures from uniformity. Finally, the coefficient of variation of binned SNP counts quantifies heterogeneity in local SNP density at a fixed genomic scale, capturing scale-dependent clustering or patchiness.

By combining these complementary metrics, the ranking distinguishes between localized, global, and scale-dependent forms of spatial heterogeneity. This provides a more robust characterization of SNP spatial structure and facilitates the interpretation of HGkI and T_{KL} results across chromosomes with different spatial properties.